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A systematic review and critical evaluation of reported pathogenic sequence variants in hidradenitis suppurativa

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Summary

Hidradenitis suppurativa (HS) is a severe chronic inflammatory disorder characterized by recurrent painful deep-seated nodules with a predilection to the apocrine-bearing areas of skin. A minority of cases of HS are due to mutations in the γ-secretase complex. Contention exists surrounding the pathogenicity of sequence variants and their effects upon Notch signalling. This systematic review was registered with PROSPERO (CRD42016041425) and was conducted in line with the PRISMA statement. Eligibility criteria for this review included published case reports, case series and reviews that identified sequence variants or protein or functional studies from patients with HS. Sixty-two articles were identified reporting a total of 41 sequence variants – heterozygous missense (nine), splice site (nine), insertion resulting in frameshift (one), premature termination codon (19) and promoter region PSTPIP1 (three) – with 18 associated protein or functional studies. The American College of Medical Genetics and Genomics standards and guidelines on the interpretation of sequence variants were applied to each identified variant to assess evidence for pathogenicity. Twenty-three variants were assessed as likely pathogenic, 17 of uncertain significance and one benign. The large number of variants of 'uncertain significance' is largely due to the variable number of functional studies. Four studies used Notch as a proxy for γ-secretase function, with conclusions of nonpathogenicity based on the assumption of Notch signalling as the sole pathogenic process. The role of Notch-independent signalling mechanisms requires further research. Limitations to this study include identification of variants of Mendelian inheritance and not complex polygenic traits.

What's already known about this topic?

- Hidradenitis suppurativa (HS) is a severe chronic inflammatory disorder, where a minority of cases are associated with sequence variants in the γ-secretase complex.
- The proposed pathogenesis for familial HS is downregulation of Notch signalling via altered function of γ-secretase.
- Conflicting results exist regarding the pathogenicity of γ-secretase variants in HS.

What does this study add?

- This systemic review has identified 41 distinct sequence variants in patients with HS.
- Applying the American College of Medical Genetics and Genomics criteria, 23 variants were likely pathogenic, 17 were of uncertain significance and one was benign.
- The luminal sites of the majority of protein alterations in the γ-secretase complex, along with lack of Notch downregulation, suggest the possibility of a Notch-independent pathway of pathogenesis in some cases of HS.

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Hidradenitis suppurativa (HS), also known as acne inversa, is a severe chronic inflammatory disorder characterized by recurrent painful deep-seated nodules with a predilection to the apocrine-bearing areas of skin.[1] A systematic review of known genetic sequence variants in HS has not previously been undertaken.

The underlying pathogenesis of HS relates to a dysregulation of the innate immune system resulting in an uncontrolled inflammatory response localized to the cutaneous follicular unit, resulting in destructive inflammation and the development of painful sinuses, abscesses and hypertrophic scarring.[1] It has strong associations with metabolic disease[2] and other inflammatory syndromes such as inflammatory bowel disease,[3,4] as well as an increased incidence in the setting of a variety of monogenic disorders[5–7] such as Down syndrome.[2] A strong autosomal dominant (AD) inheritance pattern is observed in a minority of cases of HS,[8,9] with a variable family history among the majority of individuals affected.[8,9]

Since the initial identification of a four-generation Chinese kindred with HS to chromosome 1p21.1–1q25.3 by Gao *et al.* in 2006,[8] and the subsequent identification of mutations in the γ-secretase complex by Wang *et al.* in 2010,[10] over 26 sequence variants have been reported as pathogenic variants for HS in the γ-secretase complex in multiple kindred from a variety of ethnic backgrounds.[11] The γ-secretase complex is comprised of nicastrin, presenilin 1, presenilin enhancer 2 and anterior pharynx defective 1.

The American College of Medical Genetics and Genomics (ACMG) has recently released guidelines for the stratification of pathogenicity of sequence variants for Mendelian diseases. These guidelines define objective criteria for the stratification of polymorphisms into five distinct categories: pathogenic, likely pathogenic, uncertain significance, likely benign and benign. While the criteria are applicable only for Mendelian disorders,[12] the use of objective criteria to stratify identified sequence variants in HS will aid in establishing an objective measure of pathogenicity. This may guide further investigation into proteomic studies for certain variants, as well as the search for additional monogenic or polygenic 'risk alleles' in the absence of a pathogenic variant.[12]

The importance of polygenic 'risk alleles' has become apparent as having direct relevance to disease severity and response to current therapeutic interventions, even in patients with identified monogenic variants.[13,14] Identification of these additional variants and the development of robust genotype–phenotype correlation may lead to explanations for the variable response to treatments in this disorder, as well as more accurate stratification in clinical trials and identification of more effective treatments for this highly burdensome disease.

This systematic review aims to identify all published reports of genetic polymorphisms in both familial and sporadic HS. It will also collate any protein or functional studies pertaining to these identified variants. The identified sequence variants will be critically evaluated using the ACMG standards and guidelines for the interpretation of sequence variants[12] using predefined objective criteria. This will produce a classification of sequence variants in HS to guide further functional studies, clarify the role of specific sequence variants in HS and provide insights into the pathogenesis of the disease

Methods

This systematic review was registered with PROSPERO[15] (registration number CRD42016041425) and was conducted in line with the PRISMA statement.[16] Eligibility criteria for this review included published case reports, case series and reviews that (i) included patients with clinically diagnosed HS[1] and (ii) included identified sequence variants from patients with HS or contained protein functional studies based on previously identified sequence variants from patients with HS. No restrictions on patient age, sex or ethnicity, or language restrictions were applied.

Information sources for this review encompassed MEDLINE (1946 to week 42 of 2016), Embase (1980 to week 42 of 2016) and 'Epub ahead of print, and nonindexed citations', as shown in Figure 1. The search strategy is listed in Appendix S1 (see Supporting Information).

Data collection was performed independently by two authors, with any disagreements regarding inclusion of citations being referred to a third author for mediation. Information was collected using a standardized data collection form (Appendix S2; see Supporting Information), with the principal outcome of interest being identified sequence variants.

Potential sources of bias in collating cases were acknowledged, including publications bias and reporting bias regarding the incidence of specific mutations and the rate of HS kindred with no identified mutation. Therefore no attempts were made to quantify the most common sequence variants or the proportion of kindreds without identified mutations due to this bias being present. Results were collated using the standardized data collection form. Two authors independently analysed each sequence variant using the collated data available against the ACMG guidelines (Table 1). Any disagreement was referred to a third independent author for mediation.

Results

Study selection

The results of the search strategy are illustrated in Figure 1. Of the 533 citations screened, 135 articles were identified for full-text retrieval and review. Of these 135 full-text articles, 70 were excluded due to not pertaining to gene studies or functional analysis, with three articles being duplicate reports of included cases either in abstract or review form. Sixty-two articles reported a total of 41 sequence variants, with 18 sequence variants having associated protein functional studies.

Summary of findings

The identified sequence variants and their associated protein and functional study data are summarized in Table 2. Forty-

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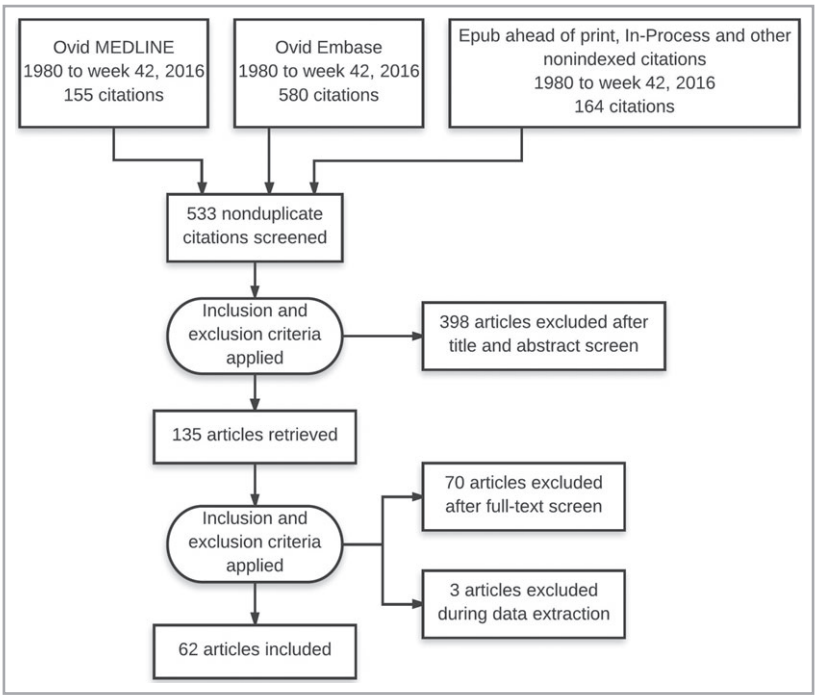

Fig 1. PRISMA flow diagram.

one individual sequence variants were identified encompassing the *NCSTN*, *PSEN1*, *PSENEN* and *PSTPIP1* genes, coding for nicastrin, presenilin 1, presenilin enhancer 2 and proline–serine–threonine phosphatase-interacting protein 1, respectively.[17–44] The types of sequence variants encompassed heterozygous missense mutations (nine variants),[17–22,28,42–44] splice-site mutations (nine variants),[10,11,22–28,30] an insertion resulting in frameshift (one variant)[11] and 19 variants resulting in premature termination codons or protein truncation.[10,11,20,26–36] Three variants[29,32] involved increased numbers of CCTG repeats the promoter region of *PSTPIP1*.

Of the 41 discrete sequence variants, 18 had associated protein or functional studies, including mRNA studies (eight variants),[10,23,26,30,32–35] transcribed protein studies (four variants),[10,20,26,30] functional studies of Notch activation (four variants)[11,17,22,38] and *in silico* predictive algorithms (two variants).[25,32,33] Twenty-seven variants were associated with familial HS,[9–11,17,19–23,25,27–30,32–35,37] six with sporadic HS[10,18,29,30,33] and one (presumed nonsignificant) with both familial and sporadic cases of HS (*NCSTN* c.1180–5C>G).[22,27] The inheritance of the remaining seven variants was not reported.[24,26,29,31]

The individual sequence variants and protein domains affected in each gene are illustrated in Figure 2. Thirty variants involved *NCSTN*,[10,11,17–37] three involved *PSEN1*[10,22,30] and three involved *PSENEN*.[10,35] All identified sequence variants in the *NCSTN* and *PSENEN* genes involved the luminal components of their respective proteins. Of the three variants in *PSEN1*, only one splice-site variant involved the cytoplasmic loop domain of the presenilin 1 protein.[22] Examining the distribution of sequence variants in *NCSTN*, no predilection could be identified pertaining to any specific protein domain or component of the nicastrin protein. Four of the six identified missense sequence variants were located adjacent to or within the Lid protein domain, which is proposed to be a vital binding site for nicastrin.[19–21,28] In fact, the only missense variant with confirmed decreased levels of Notch intracellular domain (ICD) involves a missense mutation in exon 6, within the Lid protein domain.[28] Further proteomic studies would be required to validate the hypothesis that disease-causing missense mutations are localized to this vital substrate-binding area.

A number of additional genes were reported to be identified in the setting of HS but not associated with the disease (Table 3). These included case reports of epidermolysis bullosa simplex,[38] Dowling–Degos disease[7] and keratitis–ichthyosis–deafness (KID) syndrome.[5,6] Given the lack of whole-genome sequencing in these case reports it is unknown whether any known sequence variants in the γ-secretase complex are present in these cases. Incidental association of HS with these conditions is more likely than a causative link between these diseases, unless further evidence emerges to the contrary. Evidence of cross-talk between Notch signalling and gap junction proteins does occur in embryonic development,[39,40] but it is unclear whether the association of HS and KID syndrome is more than incidental.

Reports of sequence variants that might augment the severity and onset of HS and possibly treatment response were also identified.[13,14] Haplotypes of tumour necrosis factor-α and interleukin-(IL)-12 augment the onset and severity of HS and may be confounding alleles in the pathogenicity of γ-secretase sequence variants.[13,14] Reports of individuals with HS associated with other complex polygenic disorders have identified a number of sequence variants and haplotypes that associate

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Table 1 American College of Medical Genetics and Genomics (ACMG) classification of identified genetic variants in hidradenitis suppurativa. The criteria are described in full in Table S1 (see Supporting Information)

			Evidence for 'pathogenic variant'			Evidence for 'benign variant'				
Variant	Reference	Mutation	Strong	Moderate	Supporting	Stand alone	Strong	Supporting	ACGM classification of sequence variant	Comments
1	17	NCSTN c.223G>A	–	PM2,5	PP1,4,5	–	(BS3)[a]	BP1	(Likely pathogenic)[b]	+BS3 = uncertain significance
2	28	NCSTN c.210_211delAG	PS3	PM2	PP4	–	–	–	Likely pathogenic	
3	29	NCSTN c.344_351del	PS3	PM2	PP4	–	–	–	Likely pathogenic	
4	10,23	NCSTN c.349C>T	PS3	PM2	PP4	–	BS2,4	–	Uncertain significance	
5	13	NCSTN c.477 C>A	PS3	PM2	PP4	–	(BS3)[a]	–	(Likely pathogenic)[b]	+BS3 = uncertain significance
6	30	NCSTN c.487delC	PS3	PM2,4	PP4	–	–	–	Likely pathogenic	
7	53	NCSTN c.497C>A	PS3	PM2	PP4	–	–	–	Likely pathogenic	
8	18	NCSTN c.553G>A	PS2	PM2	-	–	(BS3)[a]	BP1	(Likely pathogenic)[b]	+BS3 = uncertain significance
9	23	NCSTN c.582+1delG	PS3	PM2,4	PP4	–	–	–	Likely pathogenic	
10	19	NCSTN c.632C>G	–	PM2	PP4	–	(BS3)[a]	BP1	Uncertain significance[b]	+BS3 = likely benign
11	51	NCSTN c.647A>C	–	PM2	PP1,4,5	–	–	BP1	Uncertain significance	
12	11	NCSTN c.687insCC	–	PM2	PP4	–	–	–	Uncertain significance	
13	20	NCSTN c.887A>G	–	PM2	PP4	–	–	–	Uncertain significance	
14	21	NCSTN c.944C>T	–	PM2	PP4	–	–	–	Uncertain significance	
15	31	NCSTN c.978delG	PS3		PP4	–	–	–	Uncertain significance	
16	24	NCSTN c.996+7G>A	–	PM2	PP4	–	–	–	Uncertain significance	
17	24	NCSTN c.1101+10A>G	–	PM2	PP4	–	–	–	Uncertain significance	
18	25	NCSTN c.1101+1G>A	–	PM2	PP4	–	–	–	Uncertain significance	
19	26	NCSTN c.1125+1G>A	–	PM2	PP4	–	–	–	Uncertain significance	
20	22,27	NCSTN c.1180–5C>G	–	–	PP4	BA1	–	–	Benign	May be a polygenic risk allele
21	31–34	NCSTN c.1258C>T	PS3	PM2	PP4	–	–	–	Likely pathogenic	
22	20,30	NCSTN c.1300C>T	PS3	PM2	PP4	–	–	–	Likely pathogenic	
23	28	NCSTN c.1352+1G>A	–	PM2	PP4	–	–	–	Uncertain significance	
24	10	NCSTN c.1551+1G>A	–	PM2,4	PP4	–	–	–	Uncertain significance	
25	36	NCSTN c1635C>G	PS3	PM2	PP4	–	–	–	Likely pathogenic	
26	19	NCSTN c.1695T>G	PS3	PM2	PP4	–	–	–	Likely pathogenic	
27	10	NCSTN c.1752delG	PS3	PM2,4	PP4	–	–	–	Likely pathogenic	
28	30	NCSTN c.1768A>G	PS3	PM2,4	PP4	–	–	–	Likely pathogenic	
29	58	NCSTN c.1702C>T	PS3	PM2	PP4	–	–	–	Likely pathogenic	
30	11	NCSTN c.1799delTG	PS3	PM2	PP4	–	–	–	Likely pathogenic	
31	24,26	PSENEN c.66delG	PS3	PM2,4	PP4	–	–	–	Likely pathogenic	
32	24,26	PSENEN c.66_67insG	PS3	PM2,4	PP4	–	–	–	Likely pathogenic	
33	24,26	PSENEN c.279delC	PS3	PM2,4	PP4	–	–	–	Likely pathogenic	

(continued)

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Table 1 (continued)

Variant	Reference	Mutation	Evidence for 'pathogenic variant'			Evidence for 'benign variant'			ACGM classification of sequence variant	Comments
			Strong	Moderate	Supporting	Stand alone	Strong	Supporting		
34	10	PSEN1 c.725delC	PS3	PM2,4	PP4	–	–	–	Likely pathogenic	
35	27	PSEN1 c.837+16G>T	–	PM2	PP4	–	BS2,4	–	Uncertain significance	
36	22	PSEN1 c.953A>G	–	PM2	PP4	–	BS1	–	Uncertain significance	
37	37	PSTPIP1 CCTG5/CCTG8	–	PM2	PP4	–	–	–	Uncertain significance	
38	54	PSTPIP1	–	PM2	PP4	–	–	BP1	Uncertain significance	
39	55,56	PSTPIP1 c831G>T	PS2	PM2	PP3	–	–	–	Likely pathogenic	
40	29,55	PSTPIP1 c.1213C>T	PS2	PM2	–	–	–	BP3	Likely pathogenic	
41	29	PSTPIP1 homozygous (CCTG)5	PS2	PM2	–	–	–	BP3	Likely pathogenic	

[a]Indicates decreased Notch levels on protein studies (assuming nicastrin is functioning via the Notch intracellular domain). [b]Classification would be altered with the inclusion of BS3 criteria.

with HS (Table 3).[3–7,41–46] These variants would be appropriate for further exploration in a polygenic model of disease in individuals with sporadic HS.

American College of Medical Genetics and Genomics pathogenicity criteria analysis

The evaluation of each identified sequence variant against the ACMG criteria[12] is summarized in Table 1, with the criteria explained fully in Table S1 (see Supporting Information). Twenty-three variants were assessed as likely pathogenic,[11,17,18,20,23,28–36] 17 were of uncertain significance[11,19–22,24–26,28,31,35] and one was benign.[22,27] The one benign variant was classified as such due to the high rate of this variant in the European population (12·4%). Therefore it may be a candidate for a polygenic risk allele in future studies of sporadic HS. The classifications of sequence variants 1, 5, 8 and 10 (*NCSTN* c.223G>A, c.477 C>A, c.553G>A and c.632C>G)[17–19,25] were problematic due to assumptions made in the interpretation of Notch protein functional studies without direct evidence of protein or functional changes in the γ-secretase complex. The addition of this BS3 criterion (Table 1) is questionable as it strongly contradicts the other evidence pertaining to sequence variants 1, 5 and 8 in these highly segregated kindred.

Discussion

As well as being implicated in familial HS,[8] γ-secretase complex dysregulation is best known for its implication in early-onset and familial Alzheimer disease via Notch-independent mechanisms.[47] To date no identified sequence variants have been described as being responsible for both Alzheimer disease and HS,[47] and no increased risk of Alzheimer disease has been noted in patients or kindreds with HS.[47]

γ-Secretase dysregulation in HS has been suggested to result in downregulation of the canonical Notch signalling pathway.[48–50] This is proposed to be the 'major pathogenic mechanism of HS'.[49] Functional protein studies have produced varying results regarding the impact of γ-secretase sequence variants on Notch signalling in HS,[47,48] leading to contention regarding the pathogenicity of variants without downregulated Notch signalling.[51,52] Disparate effects upon Notch signalling have been observed in *PSEN1* sequence variants, dependent on the location of sequence variants and resultant truncation of the presenilin 1 protein.[47] Upregulation of Notch signalling was noted to occur as a result of *PSEN1* c.725delC,[47] in direct contradiction to the existing proposed pathogenesis of HS.[47–50]

Sequence variants have been identified in both sporadic and familial cases of HS,[8,9] with no single significant variant having been reported in both familial and unrelated sporadic cases.[8,9,11] This suggests variable levels of function of the resulting protein and heterogeneous impact upon Notch and Notch-independent signalling mechanisms. Well-documented HS kindreds exist with no evidence of sequence variants in the

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Table 2 Genetic sequence variants identified in this systematic review associated with hidradenitis suppurativa (HS), along with functional protein studies undertaken and familial or sporadic inheritance

Variant	Reference	Mutation	Exon/ intron	Amino acid change	Mutation type	Functional/protein studies undertaken	Proposed pathogenic effect	Associated disease	Ethnic origin	Familial/ sporadic	Notes
1	17	NCSTN c.223G>A	Ex 3	p.Val75Ile	Missense	Notch 1: no impact	NR	HS	Chinese	Familial	
2	28	NCSTN c.210_211delAG	Ex 3	p.Thr70fsX18	Truncating	Nil	NMD	HS	Chinese	Familial	
3	29	NCSTN c.344_351del	Ex 4	p.Thr115Asn*20	Truncating	Nil	NMD	PASH	NR	NR	
4	10,23	NCSTN c.349C>T	Ex 4	p.Arg117X	Truncating	↓NCTSN mRNA	PTC NMD	HS	Chinese and Japanese	Familial	
5	13	NCSTN c.477 C>A	Ex 5	p.C159X	Truncating	↓Notch 1–3/PSEN1 CTF. ↑PI3K/AKT	NMD	HS	Chinese	Familial	
6	30	NCSTN c.487delC	Ex 5	p.Gln163SerfsX39	Truncating	↓NCTSN mRNA	PTC exon 11 NMD	HS	French	Familial	
7	53	NCSTN c.497C>A	Ex 5	p. Ser166X	Truncating	Nil	PTC NMD	HS	Chinese	Familial	
8	18	NCSTN c.553G>A	Ex 5	p.Asp185Asn	Missense	Notch 1: no impact	NR	HS	British	Case	
9	23	NCSTN c.582+1delG	Int 5	Splice site	Splice site	↓NCTSN mRNA	Missing exons 6–8	HS	Japanese	Familial (2 cases)	One patient heterozygous p.Thr421Met, in PSEN2
10	19	NCSTN c.632C>G	Ex 6	p.Pro211Arg	Missense	Notch 1: no impact	Amino acid neutral to basic	HS	Chinese	Familial	
11	51	NCSTN c.647A>C	Ex 6	p.Gln216Pro	Missense	↓Notch 1–3/ICD	NR	HS	Chinese	Familial	
12	11	NCSTN c.687insCC	Ex 6	p.Cys230ProfsX31	Frameshift	Nil	NR	HS	Indian	Familial	
13	20	NCSTN c.887A>G	Ex 7	p.Pro296Arg	Missense	NR	NR	HS	Chinese	Familial	
14	21	NCSTN c.944C>T	Ex 8	p.Ala315Val	Missense	Nil	NR	HS	Chinese	Familial	
15	31	NCSTN c.978delG	Ex 8	p.M326IfsX30	Truncation	Nil	PTC NMD	HS	Chinese	NR	
16	24	NCSTN c.996+7G>A	Int 8	Splice site	Splice site	Nil	NR	HS	Mixed	NR	
17	24	NCSTN c.1101+10A>G	Int 9	Splice site	Splice site	Nil	NR	HS	Mixed	NR	
18	25	NCSTN c.1101+1G>A	Int 9	Splice site	Splice site	*In silico* studies	No effect on splicing	HS	British	Familial	
19	26	NCSTN c.1125+1G>A	Int 9	Splice site	Splice Site	↓NCTSN mRNA. GS activity normal	Skipping of exon 9		NR	NR	
20	22,27	NCSTN c.1180–5C>G	Int10	Splice site	Splice Site	NR	NR		NR	Familial (1), sporadic (2)	Allele frequency 12·4%
21	31–34	NCSTN c.1258C>T	Ex 11	p.Gln420X	Truncating	↓NCTSN mRNA	PTC NMD	HS	Chinese	Familial	
22	20,30	NCSTN c.1300C>T	Ex 11	p.Arg434X	Truncating	No reduction in NCSTN	PTC NMD	HS	French	Familial	
23	28	NCSTN c.1352+1G>A	Int 11	Splice site	Splice Site	Nil	NR	HS	Chinese	Familial	

(continued)

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Table 2 (continued)

Variant	Reference	Mutation	Exon/ intron	Amino acid change	Mutation type	Functional/protein studies undertaken	Proposed pathogenic effect	Associated disease	Ethnic origin	Familial/ sporadic	Notes
24	10	NCSTN c.1551+1G>A	Int 13	Splice site	Splice site	No reduction in NCSTN	Loss of 32 amino acids. No NMD	HS	Chinese	Familial	
25	36	NCSTN c1635C>G	Ex 15	p.Ala486 Thr517del	Truncating	NR	NMD	HS/ PASH	Iranian	Familial	
26	19	NCSTN c.1695T>G	Ex 15	p.Tyr565X	Truncating	Nil	Loss of 145 amino acids. No NMD	HS	Chinese	Familial	
27	10	NCSTN c.1752delG	Ex 15	p.Glu584AspfsX44	Truncating	↓NCTSN mRNA	NMD	HS	Chinese	Familial	
28	30	NCSTN c.1768A>G	Ex 15	p.Ser590AlafsX3	Truncating	No reduction in NCSTN	PTC exon 15 NMD	HS	French	Familial	
29	58	NCSTN c.1702C>T	Ex 15	p.Gln568X	Truncating	NR	NMD	HS	Japanese	Familial	
30	11	NCSTN c.1799delTG	Ex 16	p.Leu600X	Truncating	Nil	NMD	Acne/ HS	Indian	Familial	
31	24,26	PSENEN c.66delG	Ex 3	p.Phe23Leu fsX46	Truncating	Protein produced	Change in distal protein	HS	Chinese	Familial	
32	24,26	PSENEN c.66_67insG	Ex 3	p.Phe23ValfsX98	Truncating	↓PEN2 mRNA. Enzyme activity normal	mRNA reduced. NMD	HS	British	Familial	
33	24,26	PSENEN c.279delC	Ex 7	p.Phe94SerfsX51	Truncating	Protein produced	Change CTD PEN2	HS	Chinese	Familial	
34	10	PSEN1 c.725delC	Ex 2	p.Phe242LeufsX11	Truncating	Reduced mRNA	NMD	HS	Chinese	Three patients	↑Notch signalling in zebrafish
35	27	PSEN1 c.837+16G>T	Int 8	Splice site	Splice site	NR	NR	HS	Chinese	Case	Does not segregate
36	22	PSEN1 c.953A>G	Ex 9	p.Glu318Gly	Missense	NR	NR	HS	NR	Familial	Allele frequency 2·6%
37	37	PSTPIP1 CCTG5/ CCTG8	Promoter	Promoter	Promoter	NR	NR	HS, acne	NR	Familial	
38	54	PSTPIP1 c.831G>T		p.Glu277Asp	Missense	Nil	NR	PAPASH	NR	Case	(CCTG)6 NCSTN heterozygote
39	55,56	PSTPIP1 c.1213C>T	Ex 15	pArg405Cys	Missense	*In silico* only	NR	PASH	NR	Case	
40	29,55	PSTPIP1 homozygous (CCTG)5	Promoter	Promoter	Promoter	Nil	NR	PASH	NR	Case	
41	29	PSTPIP1 CCTG5/ CCTG8	Promoter	Promoter	Promoter	Nil	NR	PASH	NR	Case	CCTG5/CCTG8 heterozygote

CTD, C-terminal domain; CTF, C-terminal fragment; Ex, exon; GS, γ-secretase; ICD, intracellular domain; Int, intron; NMD, nonsense mediated decay; NR, not reported; PAPASH, pyogenic arthritis, pyoderma gangrenosum, acne and HS; PASH, pyoderma gangrenosum, acne and HS; PTC, premature termination codon.

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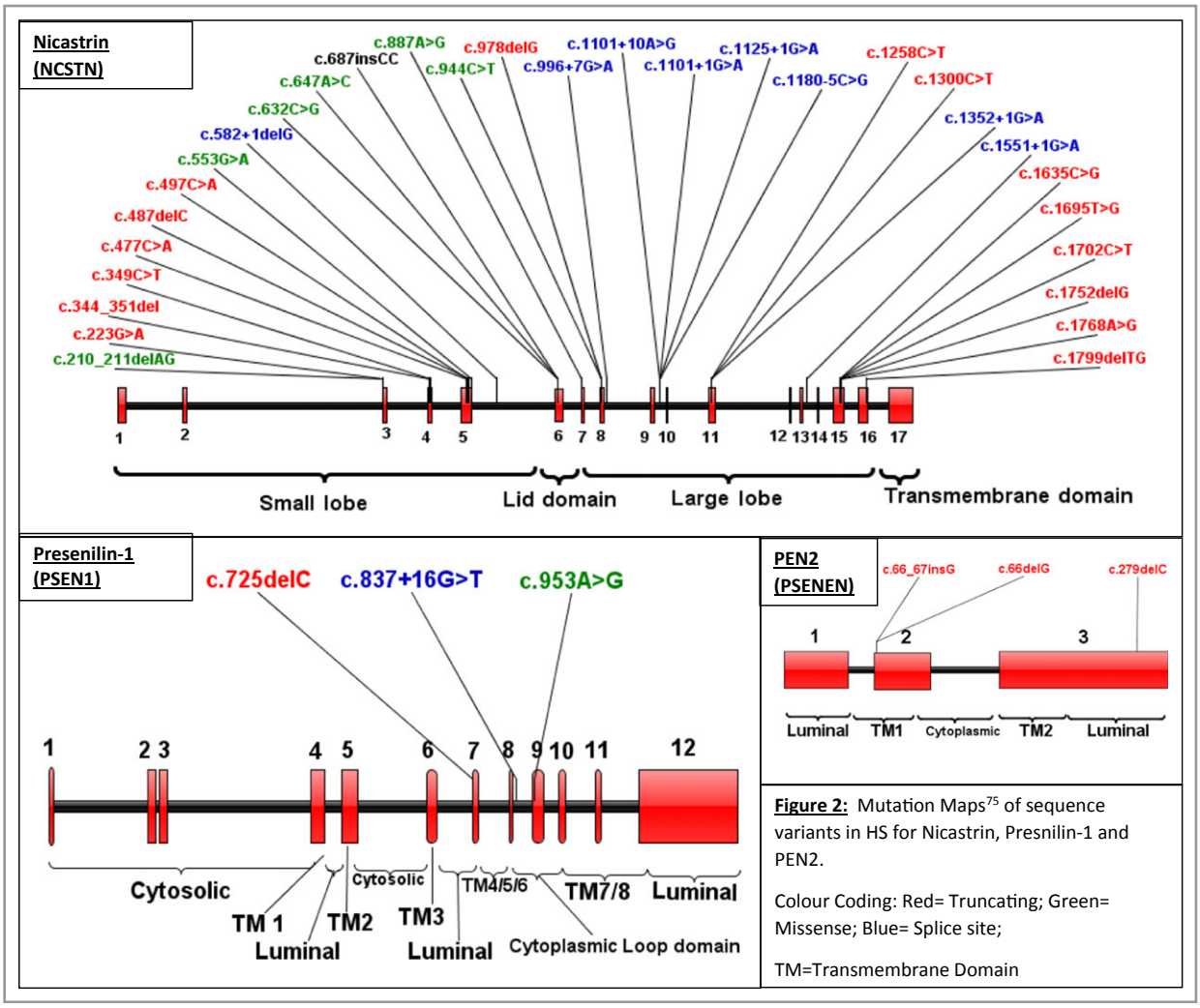

Fig 2. Mutation maps[71] of sequence variants in hidradenitis suppurativa for nicastrin, presenilin 1 and presenilin enhancer 2 (PEN2). TM, transmembrane domain.

γ-secretase complex,[10,53] suggesting involvement of additional loci in the pathogenesis of the disease. Unfortunately, functional studies of Notch signalling have not been conducted in all kindreds (with and without identified sequence variants) in order to assess whether associated loci also function through downregulation of Notch signalling. No additional genes aside from those associated with monogenic autoinflammatory syndromes (PASH/PAPASH; see below)[54–56] have been implicated to date.

The roles of environmental risk factors in the development of sporadic HS are well established[2] and have been circumstantially linked to dysregulation of Notch signalling.[49,50] All of the environmental risk factors that exacerbate or precipitate HS are proinflammatory in nature and do not exclusively impact the Notch signalling pathway.[2] From a clinical standpoint, obesity, smoking and insulin resistance are strongly related to a subset of cases of sporadic HS,[57] although some genetic predisposition to the development of HS on a background of these environmental triggers is likely at play. Circumstantially, environmental triggers are not as potent as pathogenic stimulators in familial HS,[57] although they may explain one case report of lack of segregation of sequence variants in a Japanese family.[58] The high prevalence of these environmental and metabolic triggers in the absence of HS in the general population implies additional predisposing sequence variants outside of the Notch signalling pathway to account for their strong association with sporadic HS. The possibility of a polygenic form of inheritance comprising high-frequency, low-penetrance 'risk alleles' in sporadic HS has not yet been explored.[12]

This review has identified 41 distinct sequence variants reported in association with HS, encompassing three of the four components of the γ-secretase complex (nicastrin, presenilin 1 and presenilin enhancer 2), as well as variants in the promoter region of PSTPIP1. Both HS and HS associated with PASH (pyoderma gangrenosum, acne and HS) and PAPASH (pyogenic arthritis, pyoderma gangrenosum, acne and HS) syndromes were identified as being caused by mutations in

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Table 3 Additional genes reported in association with hidradenitis suppurativa (HS)

No	Reference	Gene/sequence variant		Function of gene	Association	Comments
1	41	*SULT1B1*	NR	Conjugation of hormones, neurotransmitters, drugs	Association in the setting of HS/AI with IBD	Risk allele for HS in IBD
2	41	*SULT1E1*	NR	Oestrogen metabolism	Association in the setting of HS/AI with IBD	Risk allele for HS in IBD
3	41	*ELOVL7*	NR	Regulation of lipid metabolism via PPAR-α	Protective association in the setting of HS/AI with IBD	Protective allele for HS in IBD
4	3, 4	*NOD2* exon 4	Arg702Trp	Response to lipopolysaccharides. Implicated in Crohn disease and Blau syndrome	HS/AI and acne conglobata	Patients with HS/AI less likely to have R702W than those with Crohn disease or healthy controls
		NOD2 exon 8 *NOD2* exon 11	Gly908Arg 1007 fs		Not identified in HS/AI cohort (n = 10)	–
5	7	Keratin 5	pGly8Arg	Epidermal structural proteins	Dowling Degos and HS	Case report
6	42	Keratin 6A exon 7	1390A>C		PC type 2 and HS	Case report
7	27	Connexin 26 (*GJB2*)	p.Gly12Arg	Gap junction proteins	KID and HS/AI	Case report
	5		D50N 148G>A		KID and HS/AI	Case report
	6		C119T Ala40Val		KID and HS/AI	Case report
8	14	*IL12RB1*	D1S491 and D1S205	Cytokine	HS/AI: impact on disease onset and severity	Risk allele for disease severity
9	13	*TNF* haplotypes	Various promoter SNPs	Inflammation	Increases risk of HS/AI and increases response to tumour necrosis factor-α inhibitors	Risk allele for disease onset and treatment response
10	43	*CYP1A1*	NR	Induced by polyaromatic hydrocarbons in cigarette smoke	No elevated or decreased frequencies of *CYP1A1* activating SNP MspI or Ile462Val in HS/AI	
11	44,45	*FGFR1*	NR	Fibroblast growth factor receptor	Naevus comedonicus	Case reports
12	46	*ATP2A2*	NR	Intracellular calcium pump	Darier disease	Case report

AI, acne inversa; IBD, irritable bowel disease; KID, keratosis–ichthyosis–deafness syndrome; NR, not reported; PC, pachyonychia congenita; PPAR, peroxisome proliferator-activated receptor; SNP, single-nucleotide polymorphism.

components of the γ-secretase complex. This lends credence to the concept of HS being an autoinflammatory disorder and on a spectrum with PASH and PAPASH. PSTPIP1 and Toll-like receptor 4 are also cleaved by the γ-secretase complex (or associated downstream mediators such as ADAM10 and ADAM17) and these may be the link between γ-secretase sequence variants and the inflammasome in autoinflammatory disorders.[59]

The sites of the sequence variants in the luminal domains of the resultant proteins are in contrast to the transmembrane site of Notch ICD cleavage, although steric alterations to the transmembrane domains of these proteins cannot be confirmed without further proteomic studies. The large number of variants identified as being of 'uncertain significance' is due largely to the variable number of functional studies that have been undertaken in the setting of the identified variants and the conclusions of those studies. Identification of < 20% of the identified variants (eight of 41) involved protein or functional studies examining mRNA, or protein studies of components of the γ-secretase complex (Table 2). An additional four variants (all missense variants)[19–21,48] had accompanying protein studies looking at Notch ICD as a proxy for γ-secretase function. Only one missense variant in the Lid domain of nicastrin (*NCSTN* c.647A>C) demonstrated impaired Notch ICD signalling.[51] The authors concluded that the variants with no impairment in Notch ICD signalling were nonpathogenic.[51] This conclusion is based on the premise that Notch ICD signalling is the sole pathogenic process in HS.

Our understanding of the workings of the Notch signalling pathway is still incomplete, and evidence from Alzheimer disease research has demonstrated that sequence variants in components of the γ-secretase complex have disparate effects on

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cytosolic (Notch ICD) and luminal (extracellular) signalling, which vary according to the proteomic effect of the sequence variant.[47] Variants with no effect on Notch ICD signalling can still exert a deleterious effect on luminal signalling and vice versa.[47]

A plethora of proteins and receptor constructs are processed by the transmembrane and luminal portions of the γ-secretase complex and associated 'sheddases'.[59] This is achieved even in the absence of nicastrin protein, as nicastrin functions by steric inhibition of long extracellular domains of extracellular ligands.[60,61] Hence any abnormalities in nicastrin protein structure or function may result in dysregulation of transmembrane receptor–ligand complex cleavage, as well as dysregulation of the Notch ICD.[59–61] Such transmembrane receptors include ErbB receptors [such as epidermal growth factor receptor (EGFR) and human epidermal growth factor receptor 2 (HER2)], IL-1βR1 and IL-1βR2, IL-6R and PSTPIP precursors,[59] which have vital roles in inflammation.

In the study by Xiao *et al.*, complete absence of nicastrin (induced via small interfering RNA) does not demonstrate evidence of EGFR upregulation;[48] however, the presence of an enzymatically inactive γ-secretase complex (as measured by Notch/Hes-1 activity) does lead to upregulation of EGFR,[62] a mechanism independent of Notch signalling. Some degree of homeostatic regulation of this receptor upregulation can occur; however, this regulation is lost in the presence of a proinflammatory trigger mediated by high levels of IL-1β and IL-6.[63] Therefore, proinflammatory environmental triggers may have the potential to trigger dysregulation of γ-secretase-mediated inflammatory responses via Notch-independent pathways.

Additional evidence supporting this premise is present in studies of dysregulated cytokines in HS (such as IL-6, IL-1β and IL-17),[64] as well as reports of drug-induced HS[65–67] related to dysregulation of the ubiquitous tyrosine kinase-mediated pathways PI3K-AKT and Ras-Raf-MEK-ERK, the extracellular receptors of which undergo γ-secretase-mediated cleavage (including the EGFR and vascular endothelial growth factor receptor families).[59] In drug-induced HS due to mitogen-activated protein kinase kinase (MEK) inhibitors,[65,66] phosphoinositide 3-kinase (PI3K) activity is upregulated via release of cross-inhibition.[68] The PI3K/AKT pathway has been identified as being upregulated in HS with pathogenic mutations in *NCSTN*.[48] It is also incongruous that if Notch ICD signalling is the sole pathogenic process in HS, there are no clinical reports of HS associated with other inherited diseases involving intracellular Notch signalling[68,69] [including cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy (CADASIL) and Alagille syndrome], whereas conditions with protein precursors cleaved by γ-secretase (PASH and PAPASH)[54–56] have strong associations with HS.

Therefore it is conceivable that a proportion of cases of HS may be mediated via dysregulation of γ-secretase complex-mediated cleavage of ligands other than Notch. This would imply that a lack of abnormal Notch ICD signalling (in the setting of an otherwise likely pathogenic sequence variant in γ-secretase) does not exclude the variant from being potentially pathogenic in HS.

Evaluation of the identified sequence variants using the ACMG guidelines and recommendations is appropriate for Mendelian disorders such as familial HS.[12] Using these criteria in more complex non-Mendelian disorders can potentially lead to falsely identifying contributing sequence variants as benign. Such 'high-frequency, low-penetrance' alleles may exist in nonsporadic HS, and the identification of such risk alleles[12] is best achieved using appropriately designed linkage and association.[70] To this end we have noted in Table 2 whether the identified sequence variants are found in familial or sporadic cases of HS whenever possible *de novo* mutations may present as a case of sporadic HS. However, time and further family history may reveal an AD inheritance pattern in subsequent generations.

Further correlation is required to confirm the delineation of familial variants from those present in sporadic cases, which may be amenable to further investigation for risk alleles. It is important to note that the boundary between familial and sporadic disease is not fixed, as one Japanese kindred demonstrates,[23] where the penetrance of a given sequence variant may be altered by environmental factors such as smoking.[13] The benefit of investigating only cases and kindreds with sporadic HS for additional 'risk alleles' is that the lack of AD inheritance implies a lower penetrance of disease alleles compared with AD kindreds. Hence, comparing affected and unaffected individuals is more likely to allow accurate identification of risk alleles that can be validated in other populations.

In conclusion, this systematic review has identified 41 distinct sequence variants in HS. Of these 41 variants, 23 are considered pathogenic and 17 likely pathogenic, with only one excluded as benign. Three variants[18–20] were shown not to impact the intracellular Notch signalling pathway, bringing into question whether a certain proportion of cases may act via a Notch-independent pathway. Further proteomic and functional studies are required to correlate the pathogenic function of sequence variants in the γ-secretase complex in HS. The concept of cases of sporadic HS being a complex polygenic disorder has not been explored and is worthy of further attention given the known association of disease-altering variants in familial HS.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1 Search strategy for the systematic review.

Appendix S2 Data collection pro forma.

Table S1 Rules for American College of Medical Genetics and Genomics combining criteria.

Video S1. Author video.

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